

Jinze Li (李金泽)

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RESEARCH INTERESTS

Speculative Decoding, Agents Memory, Agent Self-evolution, LLM Post-training

EDUCATION

2023 – 2027	PhD candidate at The University of Hong Kong	Hong Kong SAR, China
2020 – 2023	Master at University of Chinese Academy of Sciences	Beijing, China
2016 – 2020	Bachelor at Dalian University of Technology	Dalian, China

INTERNSHIP

2022 – 2023	Research Intern, Huawei
2024 – 2026	Research Intern, AMD
2026 – Present	Research Intern, Ant Group (Ant Star Program)

PUBLICATIONS

- [1] **Jinze Li**, Yixing Xu, Haiduo Huang, Xuanwu Yin, Dong Li, Edith C.H. Ngai, Emad Barsoum. *Gumiho: A Hybrid Architecture to Prioritize Early Tokens in Speculative Decoding*. **ICML 2025**.
- [2] **Jinze Li**, Yixing Xu, Guanchen Li, Shuo Yang, Jinfeng Xu, Xuanwu Yin, Dong Li, Edith C.H. Ngai, Emad Barsoum. *Training-Free Loosely Speculative Decoding: Accepting Semantically Correct Drafts Beyond Exact Match*. **ICLR 2026**.
- [3] **Jinze Li**, Yang Zhang, Xin Yang, Jiayi Qu, Jinfeng Xu, Shuo Yang, Junhua Ding, Edith C.H. Ngai. *OCR-Memory: Optical Context Retrieval for Long-Horizon Agent Memory*. **ACL Main 2026**.
- [4] **Jinze Li**, Yixing Xu, Guanchen Li, Jinfeng Xu, Shuo Yang, Yang Zhang, Xuanwu Yin, Dong Li, Edith C.H. Ngai, Emad Barsoum. *Beyond the Target: From Imitation to Collaboration in Speculative Decoding*. **Submitted to ICML 2026**.

PROJECTS

- Gumiho: Hybrid Draft Architecture for Speculative Decoding (ICML 2025)**
 - Designed a hybrid draft architecture that prioritizes early tokens in the draft sequence.
 - Achieved a speedup of +4.5 to 15.8% over EAGLE-2.
 - Built the serving pipeline on vLLM for efficient, high-throughput inference.
- FLy: Training-Free Loosely Speculative Decoding (ICLR 2026)**
 - Proposed a training-free, loosely SPD method that accepts all semantically correct drafts.
 - Leveraged the target LLM’s *self-corrective* behavior to distinguish benign mismatches from truly erroneous tokens.
 - Achieved $2.81\times$ speedup on Llama-3.1-70B and $5.07\times$ on the 405B variant, outperforming EAGLE-3 by $1.62\times$ on OOD datasets.
- OCR-Memory: Optical Context Retrieval for Agent Memory (ACL 2026)**

- Proposed an agent memory framework that stores long interaction histories as images (visual tokens) to overcome text context limits without lossy summarization.
 - Achieved consistent gains on long-horizon agent benchmarks under strict context budgets: improved Mind2Web Element Accuracy to 53.8% and Step Success Rate to 46.1%.
4. **SPAD: Speculative Policy-Arbitrated Decoding (Submitted to ICML 2026)**
- Proposed a new SPD paradigm that shifts from imitation-based drafting (forcing the draft to match the target distribution) to collaboration-based decoding, which jointly leverages the strengths of both the draft and target models.
 - Trained an arbitration policy with RL in **Verl** after SFT warm-up via **LLaMA-Factory**.
 - Demonstrated substantial throughput gains: up to $3.96\times$ speedup on LLaMA-3.3-70B and $2.05\times$ on Qwen3-32B over autoregressive decoding, while leveraging draft-target complementarity to improve final-answer correctness by up to +1.65 and +2.62 accuracy points over target-only decoding, respectively.

SKILLS

1. **Proficient in vLLM**

Deployed multiple speculative decoding methods (Gumiho[1], FLY[2], Beyond-the-Target[3]) in **vLLM** for end-to-end throughput benchmarking under batched inference, familiar with vLLM's decoding pipeline.

2. **Experienced with Verl for RL**

Trained an arbitration policy model with reinforcement learning, including rollout generation, reward design, and stable policy optimization (e.g., GRPO-style training) for decoding-time decision making.

3. **Skilled in LLaMA-Factory for fine-tuning LLMs**

Used **LLaMA-Factory** to run SFT(warm-up) for the policy, then transferred the warm-started model into the RL training pipeline.

4. **Familiar with Qwen / LLaMA / DeepSeek series models**

Hands-on experience with some LLM backbones (e.g., **LLaMA-3.x**, **Qwen** series) for evaluation and optimization, and also familiar with **DeepSeek-OCR** model used in OCR-Memory[4].

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研究方向

推测解码 (Speculative Decoding), 智能体记忆 (Agents Memory), 智能体自进化 (Agent Self-evolution), LLM 后训练 (LLM Post-training)

教育背景

2023 – 2027	博士 (在读)	香港大学	中国香港
2020 – 2023	硕士	中国科学院大学	北京, 中国
2016 – 2020	本科	大连理工大学	大连, 中国

实习经历

2022 – 2023	研究型实习生, 华为
2024 – 2026	研究型实习生, AMD
2026 – Present	研究型实习生, 蚂蚁集团 (蚂蚁星计划)

论文

- [1] **Jinze Li**, Yixing Xu, Haiduo Huang, Xuanwu Yin, Dong Li, Edith C.H. Ngai, Emad Barsoum. *Gumiho: A Hybrid Architecture to Prioritize Early Tokens in Speculative Decoding*. **ICML 2025**.
- [2] **Jinze Li**, Yixing Xu, Guanchen Li, Shuo Yang, Jinfeng Xu, Xuanwu Yin, Dong Li, Edith C.H. Ngai, Emad Barsoum. *Training-Free Loosely Speculative Decoding: Accepting Semantically Correct Drafts Beyond Exact Match*. **ICLR 2026**.
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项目经历

- Gumiho: Hybrid Draft Architecture for Speculative Decoding (ICML 2025)**
 - 设计了一种混合 Draft 架构, 在 Draft 序列生成过程中优先优化靠前 token 的准确性, 以提升推理效率。
 - 相比 EAGLE-2, 实现了 +4.5–15.8% 的加速提升。
 - 基于 vLLM 搭建推理服务, 实现高吞吐、低开销的推理评估。
- FLy: Training-Free Loosely Speculative Decoding (ICLR 2026)**
 - 提出一种无需训练的 Loosely SPD 方法, 可接受所有语义正确的 Draft 输出。
 - 利用 LLM 的自我纠错 (*self-corrective*) 行为, 区分 “无害的 mismatch” 与 “真正错误的 token”。
 - 在 Llama-3.1-70B 上实现 $2.81\times$ 加速、在 405B 上实现 $5.07\times$ 加速; 在 OOD 数据集上相较 EAGLE-3 实现 $1.62\times$ 加速提升。

3. OCR-Memory: Optical Context Retrieval for Agent Memory (ACL Main 2026)

- 提出一种智能体记忆框架，将长交互历史以图像（视觉 token）的形式存储，从而在不进行有损摘要的前提下突破文本上下文长度限制。
- 在严格上下文预算下的长时程智能体基准上取得稳定提升：Mind2Web 的 Element Accuracy 提升至 53.8%，Step Success Rate 提升至 46.1%。

4. SPAD: Speculative Policy-Arbitrated Decoding (Submitted to ICML 2026)

- 提出新的 SPD 范式：从基于模仿的 Drafting (Imitation-based, 强制拟合 Target 分布) 转向基于协作的解码 (Collaboration-based)，同时利用 Draft 与 Target 模型的能力与互补性。
- 使用 LLaMA-Factory 进行 SFT 预热，并在 Ver1 中通过强化学习训练仲裁策略 (arbitration policy)。
- 取得显著吞吐提升：在 LLaMA-3.3-70B 上相对自回归解码最高加速 $3.96\times$ ，在 Qwen3-32B 上最高加速 $2.05\times$ ；同时利用 Draft-Target 互补性提升最终答案正确率，相比 target-only 解码最高提升 +1.65 与 +2.62 个百分点。

技能

1. 熟悉 vLLM

将多篇 Speculative Decoding 工作 (Gumiho[1]、FLy[2]、Beyond-the-Target[3]) 部署到 vLLM 推理框架中进行端到端吞吐与时延评测；熟悉 vLLM 的解码流程与代码结构。

2. 熟悉 Ver1 (强化学习训练)

使用强化学习训练策略模型，包含 rollout 数据生成、奖励设计，以及稳定的策略优化（如 GRPO 风格训练）。

3. 熟练掌握 LLaMA-Factory (大模型微调)

使用 LLaMA-Factory 进行 SFT (warm-up)，涵盖 LoRA 微调、数据处理、训练配置与 checkpoint 管理，并将 warm-start 权重衔接到后续 RL 训练。

4. 熟悉 Qwen / LLaMA / DeepSeek 模型

在 Speculative Decoding 场景中使用 LLaMA-3.x、Qwen 等 backbone；在 OCR-Memory[4] 中使用并熟悉 DeepSeek-OCR 作为 backbone。